SCHOOL OF COMPUTING (SOC)

CA2 Specification

Diploma in Applied AI & Analytics

ST1501
Data Engineering

2026/2027 Semester 1

Assignment rubrics

1. Demonstrate competency in data warehouse design and apply appropriate dimension modelling technique.
2. Demonstrate competency in implementing data warehouse and data ETL.
3. Demonstrate competency in implementing SQL queries to analyse data in data warehouse for trends, patterns and insights.
4. Demonstrate competency in data manipulation using NoSQL database (MongoDB).

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Section 1
Instructions and Guidelines

1. This is a **GROUP** assignment with **INDIVIDUAL** components where students are required to work in a group of 3 or 4 students to design and implement a data warehouse based on the scenario outlined in Section 2. Students must demonstrate their design through an **ERD demonstrating the appropriate dimension modelling** and implementing the data warehouse in MySQL. Additionally, students will write SQL queries to gain insights into the data such as seasonal trends and patterns. There is an individual component regarding data manipulation using NoSQL as well. Detailed tasks can be found in Section 2, and the marking scheme is outlined in Section 3.

2. This assignment accounts for **35%** of the module grade.

3. The deadline to submit this assignment is **7th August, Friday (3:59 pm).** The deadline to submit the group presentation slides is **10th August, Monday (8AM).**

4. Several files are required to complete the assignment, and they can all be downloaded from Brightspace/ST1501/CA2, where this instructional file is also located. They are:

    a. Data.zip which contains the data required. It may contain multiple files.
    b. OLTP_create.sql which contains all CREATE table statements regarding the OLTP database.
    c. CA2_group_template.docx which is the submission template to fill the answers for group tasks.
    d. CA2_individual_template.docx which is the submission template to fill the answers for individual tasks.

    Note: More information regarding the SP Policy on Use of AI Tools for Academic Work is as follows: Intellectual Property, Copyright, Plagiarism, Use of AI Tools for Academic Work. **You are supposed to submit your declaration via this link before the deadline of this assignment. Failing to do so will incur penalty.**

5. Submission should be made via the Brightspace ST1501 CA2 Assignment Submission links by the stated deadline. There will be 1 submission link for the group part and one submission link for the individual part. For group submission, only the leader of each group needs to submit. **Only the last submission is kept and graded.** Students should check and confirm their final submission is correct, complete and files are not corrupted.

6. Group deliverable for task a) to d) should be a zip file with the following file-naming convention
   "YourClass-YourGroup.zip"
   e.g. "2A01-01.zip"

   This single zip file is to be submitted to CA2 group submission link, by group leader of each group only. It should include the following items:
   a. Fully entered CA2 submission group template. You are responsible for ensuring your final submission contains your latest answer, and the file is not corrupted.
   b. Various script files containing your SQL statements, such as OLTP_insert.sql, DW_create.sql, DW_insert.sql and DW_query.sql.

   In addition, each group is required to submit their presentation slides by **8AM, 10th Aug** using the group submission link. The slides shall be submitted separately from the above ZIP file, and only the group leader should make the submission.

7. Individual deliverable for task e) should be a single zip file with the following file-naming convention
   "YourStudentID-YourName.zip"
   e.g. "2128883-JeffLim.zip"

   This single zip file is to be submitted to CA2 individual submission link. It should include the following items:
   a. Fully entered CA2 submission individual template. You are responsible for ensuring your final submission contains your latest answer, and the file is not corrupted.

8. As part of the assignment requirements, you will need to give a short presentation/demonstration to your module tutor using the submission template in your assignment group. Your module tutor may ask you questions related to your data warehouse design, implementation and SQL queries

submitted. Demonstration of executing the codes or writing SQL queries may be required. MongoDB questions may be asked as well.

9. The interview will take place during regular class time in week 17 of term 2, i.e. **11th Aug 2025 to 14th Aug 2025**. Your tutor will arrange and notify you about the detailed schedule. Every student must attend the interview and performance during the interview is graded individually. Members of the same group will attend the same interview slot together.

10. **50% of the marks will be deducted for assignments that are received within ONE (1) calendar day after the submission deadline. No marks will be given thereafter**. Exceptions to this policy will be given to students with valid LOA on medical or compassionate grounds. Students in such cases will need to inform the module tutor as soon as reasonably possible. Students are not to assume on their own that their deadline has been extended.

11. Warning: Plagiarism means passing off as one's own the ideas, works, writings, etc., which belong to another person. In accordance with this definition, you are committing plagiarism if you copy the work of another person and turn it in as your own, even if you would have the permission of that person. One shall not allow others to copy his work, and such action will result in similar penalties as plagiarism. Plagiarism is a serious offence, and if you are found to have committed, aided, and/or abetted the offence of plagiarism, disciplinary action will be taken against you. If you are guilty of plagiarism, you may get 0 marks for the assignment, fail all modules in the semester, or even be liable for expulsion. As such, the normal SP's academic policies on Copyright and Plagiarism applies. Please note that you are to cite all sources. You may refer to the citation guide available at: https://sp-sg.libguides.com/citation.

Section 2
Scope of the assignment

Note: The business scenario remains the same as described in CA1 (repeated below for your reference). However, in this CA2, the relational database (OLTP) schema is already designed. The CREATE table statements for the OLTP tables are provided in "OLTP_create.sql". You should NOT modify the statements and use them exactly to create your OLTP tables and proceed further.

Business Scenario

Green Grass Library is transitioning its loan management process from a manual, paper-based system to a fully digital platform. ~~To support this transformation, you, as a data engineer, have been tasked with designing and implementing a robust relational database system that will efficiently manage and streamline the library's operations.~~

The system is expected to maintain accurate records, enforce borrowing rules consistently, and ensure that all transactions are properly tracked. Refer to the paragraphs below regarding its operational information.

Member Registration and Management
Individuals who wish to borrow materials are required to register as members before accessing library services. During registration, the library collects essential personal information such as the member's full name, contact details, date of birth, and gender. Some additional details, such as occupation, are optional. Each member is assigned a unique identifier called member ID. The system should also record the date and time of registration.

The library operates a tier-based membership structure, which classifies members into different levels of access. There are currently three membership tiers, ranging from lowest to highest: Basic, Advanced, and Premium. All new members begin at the lowest tier upon registration. Higher tiers allow members to borrow a larger number of items at any one time. The exact borrowing allowance varies across tiers and is defined by library policy, which may change in the future. Currently, a Basic member may borrow up to 10 books at a time, an Advanced member up to 20 books, and a Premium member up to 30 books.

Membership upgrades are initiated when a member submits an application. A member will be promoted to the next tier if the following conditions are satisfied at the time of application:

- The member is at least 16 years old.
- The member has no more than 5 late returns within the past 3 years. A late return refers to a loan where the actual return date is strictly after the due date.
- The member has no more than 1 lost book record within the past 3 years.

Staff Operations

Library staff are responsible for handling daily circulation activities, including issuing and receiving borrowed items. Each staff member has a unique staff ID, and is recorded in the system with some basic personal and employment details, such as gender, employment type, start date, and assigned department.

Staff may work on either a full-time or part-time basis and are identifiable within the system for accountability in transaction processing.

Book Collection

The library maintains a collection of books covering various subjects and genres. Each book is uniquely identified by a book ID and contains descriptive information such as title, author, and genre.

The library may hold multiple physical copies of the same book. Each copy is managed individually, as they may have different availability statuses at any given time. For operational convenience, copies of the same book are distinguished using a sequence number (e.g., 1, 2, and so on) within that book ID. Each copy is tracked separately and can exist in different states depending on its current condition or circulation status, such as Available, On Loan, or Lost.

A key operational rule is that a single physical copy cannot be borrowed by more than one member at the same time. However, it is possible for a member to borrow multiple copies of the same book if available.

Certain books may have restrictions on who can borrow them. These restrictions are based on membership tier, where some books are only accessible to members who meet a minimum level of membership.

Loan and Borrowing Process

Whenever a member borrows a book, the system must record the transaction details, including who borrowed the item, which staff member handled the process, the date and time when the item was issued, the due date, and the date and time when it is returned (if applicable). Each transaction must be handled by a single staff member to complete the loan process. The due date is 14 days from the borrowing date, regardless of the book or membership tier.

If a book is reported as lost by the member, the system should record the reporting date and time as the return date and update the status of the corresponding copy to Lost.

System Objectives

The proposed system should support the following goals:

- Maintain accurate and consistent records of members, staff, and book inventory.
- Track borrowing activities and item availability over time.
- Enforce borrowing constraints based on membership rules.
- Support evaluation of membership progression requirements.
- Ensure consistency and integrity of data across all operations.
- Allow for future expansion of the membership base, library collection, and changes to membership tiers and policies.

In some cases, items may be reported as lost during a loan period, and this must be reflected in the system accordingly.

Your Tasks

Several years after the successful implementation of the operational database system, the library would like to further support business analytics. As a data engineer, you are tasked with designing a data warehouse (DW) based on the existing OLTP system to support large-scale data analytics**,** utilizing the data available in the previously described OLTP system. You need to implement the DW in MySQL, load historical data into the DW, and perform business analysis using the DW constructed.

The primary analytical focus of the data warehouse should be loan duration and borrowing activity based on **completed loan transactions (books borrowed and returned)**. Additional analyses may include transaction volume, book popularity and seasonal trends.

You will work in your assignment group to complete **Tasks a) to d)** and submit your answers using CA2_group_template.docx. **Task e) and f)** must be completed **individually** and submit your answers using CA2_individual_template.docx.

a) Set up the OLTP database in MySQL with the CREATE table statements provided in OLTP_create.sql. **No modification of the statements is allowed**. You should transform and clean the data to fit the table specifications.
   - Name your OLTP database using the format: CA2XXXXYY, where: XXXX = Your class number, YY = Your group number. For example, for group 02 in class 2A01, the correct database name would be: CA22A0102

   - Clean the data provided in data.zip before inserting them into the OLTP database tables. Record the data quality issues identified and how they were corrected in the respective table under task a) in the submission template. You may refer to topic E slides regarding possible data issues and inconsistencies to look for.
     Note that the data is guaranteed to have no business rule violation regarding loan policy:
     - At any time, no transaction violates the loan policy, such as a member borrows a book his membership tier does not permit or a member borrows a book that results in exceeding his borrow limit.
     - Refer to Topic E for some sample data quality issues.

- Data.zip consists of:
  - Membership policy information in a word document: MembershipTier.docx.
  - Library staff information in a csv file: staff.csv.
  - All other data needed including loan transaction over multiple years in data.xlsx.

- Submit your INSERT statements for data insertion in OLTP_insert.sql. If your group applies bulk loading from data files, **submit respective data files with the cleaned data as well**.

In addition, paste your database diagram for your OLTP database in the given box in submission template for task a). You can use "database/reverse engineer" function in MySQL Workbench to achieve that.

b) Design a data warehouse to support the business analytics outlined at the beginning of the **"Your Tasks"** section. Your design should address the business questions by applying appropriate dimension modelling techniques, using either a star schema or a snowflake schema. You do not need to consider slowly/fast changing dimensions in the ERD and implementation, but you can discuss about it, and you may be asked about it during interview.

   Please design your time dimension with care. Modelling time down to the second level results in an excessively large dimension table and is typically unnecessary for most analytical use cases. For this assignment, you may **limit the granularity of your time dimension to the day level** (i.e., include attributes such as date, day of week, month, year, etc.).

   Create an ERD to illustrate your data warehouse design. Paste this ERD as your answer for Task (b) in the submission template. Write a short description about your design logic, as well as additional assumptions, considerations.

c) Implement the data warehouse following your design in task b) in MySQL. Approach your tutor for feedback about your dimensional model before your implementation.

- The database name for this data warehouse is CA2DWXXXXYY, where XXXX=your class number, and YY is your group number. For example, group 01 from class 2A01 should name your DW as CA2DW2A0101.

- **Implement surrogate key for all tables, including the time dimension table, and fact table(s).**

- Create the tables for your DW. Submit DW_create.sql including the CREATE table statements for all the tables in your DW.

- Load data from the OLTP table and insert into your DW using appropriate SQL statements. Submit all your insert statements with the data loading part in DW_insert.sql.

In addition, paste your database diagram for your data warehouse in the given box in submission template for task c). You can use "database/reverse engineer" function in MySQL Workbench to achieve that.

Briefly explain how you verified the accuracy of your data loading. Support your explanation by referencing relevant facts or summary queries that demonstrate the data has been correctly extracted, transformed, and loaded into your data warehouse.

d) Provide a set of meaningful analytical queries that your data warehouse can support to extract insights from the available historical data. These queries should address the types of business questions mentioned earlier.

Submit your SQL queries in a file named DW_query.sql. For each query, provide a brief explanation of what it does, the result of the query, and a short discussion of the findings and insights derived under task d) in the submission template.

Below are some suggested areas of analysis you may wish to explore in your queries. You are also encouraged to investigate other aspects of the data that could offer valuable insights. If you plan to explore areas beyond those listed, please consult your tutor to ensure they are relevant to the assignment scope.

Suggested Analytical Questions:

- Are there any **trends or patterns** in loan transactions over time (e.g., by month, day of week)?
- Are there any **outstanding members** with significantly higher/lower total transaction/loan duration?
- Are any book genres more popular than others?
- Any other analysis involving more than one dimension?

Your group is expected to demonstrate 6 analytics if your group size is 3 members, or 8 analytics if your group size is 4 members. i.e. Every member must contribute two substantially different analytical queries. Your analytics should cover at least three different aspects (in terms of measurements or dimensions) of the business data. You are encouraged to explore insights that involve long-term/seasonal trends or analyses across multiple dimensions.

e) **This is an individual task**. Follow the instructions to build one collection, insert data into them from the **OLTP** tables, and write MongoDB queries as required. All your tasks should be completed using MongoDB Shell (mongosh) commands only. You should not use any Mongo GUI or other languages.

i. Data import from OLTP and MongoDB collection creation
With reference to the OLTP database created and populated in task a), create a Mongo database called CA2Mongo with 1 collection:

1. A Book collection that stores one document for each book. Each document should include the book ID, title, author, genre, minimum membership tier required to borrow the book, total number of physical copies, number of lost copies, and total number of loan transactions recorded for that book (sum among all copies given the data available).

Create the collection in MongoDB and insert data by extracting the required information from the **OLTP** tables. Write and paste the SQL query used to extract the data under Task e)(i) of the submission template. In addition, fill in the count of records in the collection on the table provided in the submission template.

ii. Use MongoDB commands to answer the following questions by querying your collection created in part i). For each question, paste your MongoDB command and the results in the submission template.

1) List the titles, genres, and loan counts of all books that have been borrowed more than 400 times. Display the results in descending order of loan count.

2) For each genre, display the number of books and the average number of loans per book. Sort the results by average number of loans in descending order. Show only those genres with more than 50 different books, and average number of loans per book greater than 70.

f) **This is an individual task**. You are required to explore a single topic related to data engineering that extends beyond the scope and depth covered in this module. Examples of suitable topics include advanced SQL techniques, indexing and query optimisation, database performance tuning, alternative data warehouse modelling approaches, NoSQL database internals, data lake architectures, or other relevant data engineering topics.

Your report should focus on a specific aspect of the chosen topic rather than providing a broad overview. For example, instead of discussing NoSQL databases in general, you may explore how document databases can be used to model library loan transactions or investigate how a particular indexing strategy affects query performance.

During your study, you may refer to any appropriate resources, including tutorials, videos, technical articles, documentation, sample code, academic papers, or AI tools. If AI tools are used, clearly state how they were used and how they contributed to your learning, understanding, experimentation, or report preparation.

You should submit a report of no more than 5 pages using a readable 12-point font. The report should include:

- The question, problem, or topic being explored.
- The key concepts, techniques, or findings learned.

- A discussion of how the topic can be applied to a relevant scenario.
- A practical demonstration using sample data, queries, code, system outputs, or other appropriate evidence.

The report shall be included in the Task f) section of the individual submission template. You may use the data provided in this assignment, generate your own sample data, or use a combination of both for your demonstration.

The application scenario should be related to library operations or library transaction data. However, the detailed business rules, assumptions, and data used in your scenario may differ from those stated in this assignment.

Refer to the marking scheme for the detailed assessment criteria for this task.

Section 3
Marking Scheme

Marks will be awarded to each student based on the following rubrics:

Component	Task	Weight
Group	**a) – Setting up the OLTP Tables** • Tables are created exactly using the statements provided. • Data quality is sufficiently improved. • Cleaned data is inserted into the tables with no error. • Demonstrate the correctness of OLTP setup.	10%
	b) – Data Warehouse Design • The design is in the appropriate form of a star or snowflake schema, which supports the business analytics required in section 2. • The chosen table names, field names and attributes are descriptive. • The explanation of the design is clear and concise.	15%
	c) – Create the Data Warehouse • Create the DW tables with correct Primary key, surrogate key, and foreign keys. • DW tables are populated with data correctly.	15%
	d) – Data Warehouse Analytics • The query objective is meaningful. • The query and result are correct. • The discussion is insightful. • Cover at least 3 different business aspects.	20%
Individual	**e) i – MongoDB Setup** • The collection and documents are set up correctly. • The data are queried correctly from OLTP tables. • The number of records in each collection is correct.	5%
	e) ii – MongoDB Queries • The query statements and results are correct.	10%
	f) Individual report • Selected topic is meaningful and not trivial. • Discussion is adequate and correct. • Demonstrate the discussion using sample data and queries.	15%

CA2 Assignment Demo/Interview The CA2 Demo/Interview will be conducted where you have: • To explain/demonstrate the data warehouse design and other deliverables. • To explain/demonstrate your MongoDB queries. • To explain/demonstrate the topics you explored in f). • To explain/answer questions given by the module tutor. • All members of a group need to attend the interview together in designated slot. The grading is individual.	10%
Total	100%

-- End of Assignment Specifications --